



UNIVERSITY
of NICOSIA | DEPARTMENT OF
COMPUTER SCIENCE

University of Nicosia, Cyprus COMP-358 Networks and Data Communication

Project II for the course Week 12

Project Title:

Exploration of Network Modes and secure Multi-Device communication and sync in Communication Apps

Purpose

To evaluate how communication apps like Teams handle audio/video connections and multi-device synchronization under different network scenarios, identifying key challenges and recommending best practices for reliability and security.

Important note: The project is due on the day of the final examination, specifically 2 hours before the scheduled start time of the exam.

Case Studies/Scenarios:

(A) When an application like Teams (or a similar one) establishes an audio channel for telephony calls, it can do so in three ways/forms/modes:

- 1) Direct connection using UDP.
- 2) Indirect connection using UDP, with selective acknowledgements relayed via a Supernode.
- 3) Indirect connection using TCP to reach a Supernode, then UDP from the Supernode to the destination, with selective acknowledgements.

Based on your understanding and the theoretical considerations of the modes described above, please answer the following questions:

- a) Why does Teams (app or Desktop application) provide these three modes? Your answer should be aligned in terms of the operations that are performed on the upper-level layers (software layers) starting from the transport layer.
- b) Describe the different audio/video problems you might encounter when the (1) and (3) modes are used.
- c) Which mode will normally provide the best audio/video experience? Why?
- d) Assume that there is an asynchronous transfer mode, in a selective acknowledgement way of providing positive and negative acknowledgements. How do you think the Teams (app or Desktop application) will operate in terms of reliability, delay variations, interarrival delays between the different packets of the same stream etc.? Would the ATM be the best case for having Teams connectivity? Substantiate your answer.

(B) Scenario Description:

Imagine Device A, which is a mobile device that is connected to a VPN server. This VPN encrypts the internet traffic of Device A, ensuring secure communication. Now, there is another device (let's call it Device B) that remotely tries to synchronize its content with Device A (not through a VPN). Both devices are logged into the same account of an app (e.g., Viber, Microsoft Teams, WhatsApp). The synchronization is initiated with the consent of the owner.

Based on the above answer the following questions with substantiated answers on Issues and technical challenges:

1. How does using a VPN affect synchronization between devices in terms of performance and reliability? Do you think there will be compatibility issues with this synchronization?
2. How do these apps in your view and the knowledge that you gained handle multi-device synchronization while maintaining reliability and security?
3. What role does end-to-end encryption play in ensuring secure communication between synchronized devices?
4. If Device B connects from a different country or IP address than Device A, how might this impact the app's security mechanisms?
5. What steps can developers take to ensure robust reliability and security when designing multi-device synchronization features?
6. What are some limitations of using a VPN for securing communications in multi-device setups?

Your answers (maximum 200 words for each question 1-6) should be substantiated based on each question and the scenario described above.